

MANEET CHATTERJEE

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SUMMARY

Mechanical Engineering researcher working in AI systems for aerospace, robotics, and autonomous engineering with R&D experience across Synopsys, Ansys, and academic research spanning generative design, embodied AI, digital twins, computer vision, physics-based simulation, and rigorous model validation.

EDUCATION

Indian Institute of Engineering Science and Technology Shibpur

Aug 2023–May 2027

B.Tech. in Mechanical Engineering | CGPA: **3.63/4.00** (through Semester VI)

TECHNICAL SKILLS

Programming & ML: Python, C++, MATLAB, SQL, JavaScript, Julia; PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Pandas

Robotics & Systems: ROS 1/2, NVIDIA Isaac Sim, Gazebo, MuJoCo, PyBullet, Simulink, Raspberry Pi, MQTT, Git, Docker

CAD & CAE: Siemens Designcenter, CATIA, SolidWorks, Fusion 360, AutoCAD; Ansys Mechanical, Abaqus

CFD, Aerospace & Multiphysics: SUAVE, OpenVSP, VSPAERO, Ansys Fluent, STAR-CCM+, OpenFOAM, COMSOL Multiphysics

Digital Engineering & Data: Digital Twins, IT-OT Integration, InfluxDB, Grafana, Microsoft Excel, Power BI, LaTeX

EXPERIENCE

R&D Intern | Synopsys Inc., Pune, India

Jun 2026–Present

- Developed a generative inverse-design framework for conceptual eVTOLs using a canonical aircraft schema encoding components, configurations, topology, and placement rules; coupled component-aware sampling to SUAVE mission analysis, native OpenVSP geometry, and VSPAERO aerodynamics.
- Ran conditional latent diffusion, with **91%** of generated designs passing geometric and forward-model checks; logged solver settings, convergence, engineering validity, and provenance, then extended the workflow to Formula SAE and gem5/McPAT PPA studies. [Certificate](#)

Research Assistant | University of Tsukuba, Tsukuba, Japan

Mar 2026–Present

- Architected a MedSAM/SAM 3 pipeline across **28 laparoscopic procedures, 142 annotated events, and 14,776 frames**, localizing linear/circular stapler insertion, compression, firing, pin penetration, closure, and removal.
- Fused spatial/frame features, VideoMAE clips, stapler-region motion, and hard negatives; achieved **91.6% balanced accuracy** and **89.7% F1** under five-fold case-disjoint CV with mixed-precision distributed training, caching, calibration, and safety-rubric assessment.

R&D Intern | Ansys Inc., Pune, India

May 2025–Jul 2025

- Developed robotic systems integrating embedded controllers and sensors with MQTT/WebSocket remote monitoring and control; deployed secure communication and local data management on Raspberry Pi.
- Built OpenCV/PyTorch/ArUco vision-guided manipulation with object detection, calibration, coordinate transforms, and robot kinematics; validated behavior in Isaac Sim, Blender, and Unreal Engine. [Certificate](#)

Research Assistant | IIST Shibpur, Kolkata, India

Aug 2024–Dec 2024

- Modeled solar-powered hydrogen production from measured irradiance/weather data, optimizing PV–electrolyzer power matching at **97% electrolyzer efficiency**. [Report](#)

SELECTED PROJECTS

Physics-Gated Generative CAD for Aircraft Inverse Design

Sept 2026

- Engineered a Python/SUAVE inverse-design pipeline spanning **43,702 aircraft classes and 17 energy networks** using probabilistic grammar, Sobol sampling, **30 measured physical constraints**, and automated 3D CAD generation.
- Integrated converged SUAVE mission analysis, conservation-law validation, and watertight STL/OBJ export; delivered **21/21 physically valid, watertight CAD designs** and **11/11 validated turbofan solutions**, with climb fuel optimized to **10.2% of MTOW**.

CFD Validation of Conjugate Heat Transfer in Curved Cooling Ducts

May 2026

- Cross-validated matched curved-duct cases in Fluent, COMSOL, and OpenFOAM by solving steady incompressible Navier–Stokes and energy equations with $k-\omega$ SST modeling; identified Dean-type vortices governing thermal non-uniformity.
- Improved mean Nusselt number by **18%** over the project baseline while holding pressure-drop predictions within **6%** across solvers, supporting EV-battery, avionics, and aerospace-ECS cooling design.

Vision-Language-Action Control for Robotic Manipulation

[GitHub](#)

- Built a CLIP ViT-B/32-guided PyBullet pick-and-place pipeline with object-level perception for **5 objects and 4 sorting zones**.
- Defined a target-vs.-distractor confidence margin to quantify semantic ambiguity; separated successful and failed executions (Mann–Whitney $p < 0.01$, ROC-AUC = 0.75).

MoonBot Design and Reinforcement Learning

[GitHub](#)

- Designed a **6-wheel rover with an articulated arm** in SolidWorks; converted it to URDF with inertial properties, joint constraints, and collision geometry for stable multi-contact dynamics in PyBullet/MuJoCo.
- Built a Gym-compatible environment and trained PPO locomotion with mass/friction domain randomization for robust sim-to-real transfer using Gymnasium and Stable-Baselines3.

SELECTED PUBLICATIONS

- M. Chatterjee, A. Ghosh, A. Sen, U. Sen.** *Topology-Constrained Graph-Mamba with Logit Diffusion for Change Detection*. **MONTI Workshop at CVPR, 2026. First author; non-archival workshop.**
- A. Roy, A. Sen, **M. Chatterjee**, S. Das, J. Paul. *Beyond CNNs: EfficientMamba Powered Sequence Learning and Curriculum-Guided Domain Adaptation for Fossil Classification*. **IEEE InGARSS, 2025.**
- M. Chatterjee**, A. Sen, S. Roy. *EEMAMBA: A Hardware-Aware Energy-Efficient State-Space Model for EuroSAT Classification*. **IEEE IGARSS, 2025. First author. DOI**
- M. Chatterjee**, A. Sen, S. Roy. *ExoSpikeNet: A Light Curve Analysis Based Spiking Neural Network for Exoplanet Detection*. **IEEE CSNT, 2024. First author. DOI**

HONORS, PRESENTATION & CERTIFICATIONS

Honors: GAABESU Research Award (2025); IEEE GRSS Travel Grant for IGARSS (2025). **Presentation:** Co-authored/presented an IT-OT and digital-twin paper at the 6th Ansys TechCon & 2nd AIIoT (2025).

Certifications: Deep Learning Specialization (2026); [SOLIDWORKS Professional](#) (2025); [MATLAB Deep Learning](#) (2024).